

Unit 38: Trigonometric Integrals — Full Solutions

Warm-up

1. $\int \sin^3 x \, dx$

Odd power on sine. Write $\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x) \sin x$. Let $u = \cos x$, $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$.

$$\int (1 - u^2)(-du) = -\int (1 - u^2) du = -u + \frac{u^3}{3} + C = -\cos x + \frac{\cos^3 x}{3} + C$$

2. $\int \cos^3 x \, dx$

Odd power on cosine. Write $\cos^3 x = (1 - \sin^2 x) \cos x$. Let $u = \sin x$, $du = \cos x \, dx$.

$$\int (1 - u^2) du = u - \frac{u^3}{3} + C = \sin x - \frac{\sin^3 x}{3} + C$$

3. $\int \sin x \cos^2 x \, dx$

Let $u = \cos x$, $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$.

$$\int u^2(-du) = -\frac{u^3}{3} + C = -\frac{\cos^3 x}{3} + C$$

4. $\int \sin^2 x \cos x \, dx$

Let $u = \sin x$, $du = \cos x \, dx$.

$$\int u^2 du = \frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C$$

5. $\int \cos^3 x \sin x \, dx$

Let $u = \cos x$, $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$.

$$\int u^3(-du) = -\frac{u^4}{4} + C = -\frac{\cos^4 x}{4} + C$$

6. $\int \sin^2 x \, dx$

Power-reducing identity:

$$\int \frac{1 - \cos(2x)}{2} dx = \frac{1}{2} \left[x - \frac{1}{2} \sin(2x) \right] + C = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

Standard

7. $\int \cos^2 x \, dx$

$$\int \frac{1 + \cos(2x)}{2} \, dx = \frac{1}{2} \left[x + \frac{1}{2} \sin(2x) \right] + C = \frac{x}{2} + \frac{\sin(2x)}{4} + C$$

8. $\int \sin^3 x \cos^2 x \, dx$

Odd power on sine. Write $\sin^3 x = (1 - \cos^2 x) \sin x$.

$$\int (1 - \cos^2 x) \sin x \cos^2 x \, dx = \int (\cos^2 x - \cos^4 x) \sin x \, dx$$

Let $u = \cos x$, $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$.

$$\int (u^2 - u^4)(-du) = -\left[\frac{u^3}{3} - \frac{u^5}{5} \right] + C = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

9. $\int \cos^3 x \sin^4 x \, dx$

Odd power on cosine. Write $\cos^3 x = (1 - \sin^2 x) \cos x$.

$$\int (1 - \sin^2 x) \cos x \sin^4 x \, dx = \int (\sin^4 x - \sin^6 x) \cos x \, dx$$

Let $u = \sin x$, $du = \cos x \, dx$.

$$\int (u^4 - u^6) \, du = \frac{u^5}{5} - \frac{u^7}{7} + C = \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C$$

10. $\int \sin^4 x \cos x \, dx$

Let $u = \sin x$, $du = \cos x \, dx$.

$$\int u^4 \, du = \frac{u^5}{5} + C = \frac{\sin^5 x}{5} + C$$

11. $\int \cos^5 x \, dx$

Write $\cos^5 x = (\cos^2 x)^2 \cos x = (1 - \sin^2 x)^2 \cos x$. Let $u = \sin x$, $du = \cos x \, dx$.

$$(1 - u^2)^2 = 1 - 2u^2 + u^4$$

$$\int (1 - 2u^2 + u^4) du = u - \frac{2u^3}{3} + \frac{u^5}{5} + C = \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$$

12. $\int \sin^4 x dx$

Both powers even. Write $\sin^4 x = (\sin^2 x)^2 = \left(\frac{1 - \cos(2x)}{2} \right)^2 = \frac{1 - 2\cos(2x) + \cos^2(2x)}{4}$.

Reduce the leftover $\cos^2(2x)$ term using the identity again: $\cos^2(2x) = \frac{1 + \cos(4x)}{2}$.

$$\sin^4 x = \frac{1 - 2\cos(2x) + \frac{1 + \cos(4x)}{2}}{4} = \frac{3 - 4\cos(2x) + \cos(4x)}{8}$$

$$\begin{aligned} \int \sin^4 x dx &= \frac{1}{8} \int [3 - 4\cos(2x) + \cos(4x)] dx = \frac{1}{8} \left[3x - 2\sin(2x) + \frac{1}{4}\sin(4x) \right] + C \\ &= \frac{3x}{8} - \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C \end{aligned}$$

Challenge

13. $\int \sin^5 x dx$

Write $\sin^5 x = (\sin^2 x)^2 \sin x = (1 - \cos^2 x)^2 \sin x$. Let $u = \cos x$, $du = -\sin x dx \Rightarrow \sin x dx = -du$.

$$(1 - u^2)^2 = 1 - 2u^2 + u^4$$

$$\int (1 - 2u^2 + u^4)(-du) = - \left[u - \frac{2u^3}{3} + \frac{u^5}{5} \right] + C = -\cos x + \frac{2}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C$$

14. $\int \sin^6 x \cos^3 x dx$

Odd power on cosine. Write $\cos^3 x = (1 - \sin^2 x) \cos x$.

$$\int \sin^6 x (1 - \sin^2 x) \cos x dx = \int (\sin^6 x - \sin^8 x) \cos x dx$$

Let $u = \sin x$, $du = \cos x dx$.

$$\int (u^6 - u^8) du = \frac{u^7}{7} - \frac{u^9}{9} + C = \frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + C$$

15. $\int \cos^2 x \sin^2 x \, dx$

Both even. Use the identity $\sin x \cos x = \frac{1}{2} \sin(2x)$, so $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2(2x)$.

$$\int \frac{1}{4} \sin^2(2x) \, dx = \frac{1}{4} \int \sin^2(2x) \, dx$$

Apply the power-reducing identity to $\sin^2(2x) = \frac{1 - \cos(4x)}{2}$:

$$\begin{aligned} \frac{1}{4} \int \frac{1 - \cos(4x)}{2} \, dx &= \frac{1}{8} \int [1 - \cos(4x)] \, dx = \frac{1}{8} \left[x - \frac{1}{4} \sin(4x) \right] + C \\ &= \frac{x}{8} - \frac{\sin(4x)}{32} + C \end{aligned}$$

16. $\int_0^{\pi/2} \sqrt{1 + \cos(2x)} \, dx$

Using the identity $1 + \cos(2x) = 2\cos^2 x$:

$$\sqrt{1 + \cos(2x)} = \sqrt{2\cos^2 x} = \sqrt{2} |\cos x|$$

On $[0, \frac{\pi}{2}]$, $\cos x \geq 0$, so $|\cos x| = \cos x$.

$$\int_0^{\pi/2} \sqrt{2} \cos x \, dx = \sqrt{2} [\sin x]_0^{\pi/2} = \sqrt{2}(1 - 0) = \sqrt{2}$$

Answer: $\sqrt{2}$.

17. $\int_{-\pi/4}^{\pi/4} \sqrt{1 - \cos(2x)} \, dx$

Using the identity $1 - \cos(2x) = 2\sin^2 x$:

$$\sqrt{1 - \cos(2x)} = \sqrt{2} |\sin x|$$

On $[-\frac{\pi}{4}, \frac{\pi}{4}]$, $\sin x$ is negative on $[-\frac{\pi}{4}, 0)$ and positive on $(0, \frac{\pi}{4}]$ — so we must split the integral at $x = 0$.

$$\int_{-\pi/4}^0 \sqrt{2}(-\sin x) \, dx + \int_0^{\pi/4} \sqrt{2} \sin x \, dx$$

First piece: $\sqrt{2} [\cos x]_{-\pi/4}^0 = \sqrt{2} (\cos 0 - \cos(-\frac{\pi}{4})) = \sqrt{2} (1 - \frac{\sqrt{2}}{2}) = \sqrt{2} - 1$.

Second piece: $\sqrt{2} \left[-\cos x \right]_0^{\pi/4} = \sqrt{2} (-\cos \frac{\pi}{4} - (-\cos 0)) = \sqrt{2} \left(-\frac{\sqrt{2}}{2} + 1 \right) = \sqrt{2} - 1.$

$$\text{Total} = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2\sqrt{2} - 2$$

Answer: $2\sqrt{2} - 2 \approx 0.828$. (You can double-check this using symmetry: since $\sqrt{1 - \cos(2x)}$ is an even function of x , the integral equals twice the integral from 0 to $\frac{\pi}{4}$, which is $2(\sqrt{2} - 1) = 2\sqrt{2} - 2$ — matching exactly.)

Applied

18. $\int_0^{2\pi} \sin^2 x \, dx$

Using the antiderivative from Problem 6: $\frac{x}{2} - \frac{\sin(2x)}{4}.$

$$\left[\frac{x}{2} - \frac{\sin(2x)}{4} \right]_0^{2\pi} = \left(\pi - \frac{\sin(4\pi)}{4} \right) - (0 - 0) = \pi - 0 = \pi$$

Answer: π .

19. $\int_0^{\pi} \cos^2 x \sin x \, dx$

Let $u = \cos x$, $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$. When $x = 0$, $u = 1$; when $x = \pi$, $u = -1$.

$$\int_1^{-1} u^2 (-du) = \int_{-1}^1 u^2 \, du = \left[\frac{u^3}{3} \right]_{-1}^1 = \frac{1}{3} - \left(-\frac{1}{3} \right) = \frac{2}{3}$$

Answer: $\frac{2}{3}.$

20. $\int_0^{\pi} \sin^3 x \, dx$

Using the antiderivative from Problem 1: $-\cos x + \frac{\cos^3 x}{3}.$

$$\begin{aligned} \left[-\cos x + \frac{\cos^3 x}{3} \right]_0^{\pi} &= \left(-\cos \pi + \frac{\cos^3 \pi}{3} \right) - \left(-\cos 0 + \frac{\cos^3 0}{3} \right) \\ &= \left(1 + \frac{-1}{3} \right) - \left(-1 + \frac{1}{3} \right) = \frac{2}{3} - \left(-\frac{2}{3} \right) = \frac{4}{3} \end{aligned}$$

Answer: $\frac{4}{3}.$

21. Average value $\frac{1}{\pi} \int_0^{\pi} \sin^2 x \cos^2 x \, dx$

Using the antiderivative from Problem 15: $\frac{x}{8} - \frac{\sin(4x)}{32}$.

$$\int_0^\pi \sin^2 x \cos^2 x \, dx = \left[\frac{x}{8} - \frac{\sin(4x)}{32} \right]_0^\pi = \left(\frac{\pi}{8} - \frac{\sin(4\pi)}{32} \right) - 0 = \frac{\pi}{8}$$

Average value:

$$\frac{1}{\pi} \cdot \frac{\pi}{8} = \frac{1}{8}$$

Answer: $\frac{1}{8}$.